

Akshat Sharma

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Objective

Undergraduate researcher specializing in inorganic/materials chemistry and theoretical chemistry with hands-on expertise in solvothermal synthesis, crystal structure refinement (Olex2), DFT studies (ORCA), and topological analysis (Multiwfn). Focused on the design and structural characterization of novel catalytic materials for sustainable energy and CO₂ reduction. Additionally dedicated to the theoretical analysis of molecular behavior and complexity.

Education

- **BS-MS Dual Degree:** Indian Institute of Science Education and Research Kolkata 2023–2028
- **Class XII (CBSE):** Baluni Public School, Dehradun 88.8% (2023)
- **Class X (CBSE):** Baluni Public School, Dehradun 88.8% (2021)

National Level Exams Qualified

- Joint Entrance Examination (Advanced)
- Joint Entrance Examination (Mains)
- IISER Aptitude Test (IAT)

Awards and Honors

- **Reliance Foundation Scholarship**
Selected among the top 5,000 students across India for academic excellence.

Experience

Undergraduate Researcher

Dec 2023 – Present

Supervisor: Dr. Soumyajit Roy, IISER Kolkata

- Synthesized the giant polyoxometalate cluster {Mo₁₃₂} keplerates and {Mo₁₅₄} giant blue ring using self-assembly in aqueous medium and confirmed the formation using powder X-ray diffraction, UV-vis spectroscopy, and Raman spectroscopy.
- Conducted a detailed study of the polymorphism and solid-state phase transitions of acetaminophen (paracetamol).
- Synthesized metastable Paracetamol Form II (orthorhombic) and characterized the phase transformation using Single Crystal XRD.
- Synthesized soft-oxometalates for CO₂ reduction and water oxidation catalysis.
- Analyzed crystal formation as a function of the crystallization medium and temperature by systematically varying ibuprofen concentration, DMSO content, and thermal conditions.
- Trained in advanced crystallographic refinement protocols (Müller lineage standards).

Research Intern

May 2025 – July 2025

Supervisor: Dr. C.M. Nagaraja, IIT Ropar

- Gained foundational knowledge of Metal–Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs) through an extensive review of the scientific literature.
- Synthesized sterically demanding N-heterocyclic carbene (NHC) ligands and their organometallic complexes, confirming structural integrity via ¹H NMR spectroscopy.

Publications & Preprints

- A. Sharma "Quantum assembly theory defines molecular complexity from electron density topology." **ChemRxiv** (2026) [Link]

Skills

- **Crystallography:** Structure solution and refinement (Olex2). Data reduction and unit cell determination using CrysAlisPro; packing analysis with Mercury.
- **Spectroscopy:** Operation and analysis of UV-Vis, FT-IR, and Raman spectroscopy. Data analysis of Powder XRD (Phase identification) and Single Crystal XRD.
- **Computational Chemistry:** DFT geometry optimization and thermodynamic analysis (Enthalpy, Gibbs Free Energy) using ORCA. Topological Analysis using Multiwfn. Input generation (Avogadro) and data plotting (Origin).
- **Synthesis:** Organic ligand precursors (Imidazolium salts), Solvothermal synthesis (POMs).
- **Programming and Tools:** Python, C, LaTeX, Linux.

Relevant Courses

- **Inorganic and Materials:** Coordination Chemistry, Molecular Symmetry and Character Tables, Main Group Elements.
- **Physical and Analytical:** Phase Equilibrium Thermodynamics, Quantum Chemistry, Fundamentals of Spectroscopy (IR/Raman/Electronic), Instrumentation in Chemistry.
- **Mathematics and CS:** Linear Algebra, Group Theory, Probability and Statistics, Data Structures and Algorithms (DSA), Programming in Python.